

ABSTRACT

Aquaculture plays an important role in achieving global food security, especially considering the projected population increase to over 9.8 billion by 2050. Traditional harvesting methods have exceeded ecological limits, making aquaculture essential for meeting growing food demands while preserving marine biodiversity. However, effective aquaculture management requires comprehensive ecosystem monitoring and prediction, with water quality (WQ) being a critical factor influencing production and product quality. Traditional water quality monitoring (WQM) methods are expensive and time-consuming, posing challenges for small-scale farms. The advent of technologies such as the Internet of Things (IoT), artificial intelligence (AI), and data analysis has introduced smart aquaculture systems. Nevertheless, accurate water quality prediction (WQP) remains a challenge due to the complex interdependencies and non-linear characteristics of WQ parameters. To address these challenges, this research work investigates the application of deep learning (DL) models to propose cost-effective solutions for WQM and precise WQP in aquaculture. DL models, including long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN), excel at capturing the non-linear relationships present in aquaculture WQ parameters. The primary objective is to enhance prediction accuracy while reducing computation time, enabling efficient WQ management in aquaculture. Additionally, the proposed models strive to offer economically viable solutions, making advanced technology in WQ management systems accessible and affordable.

This research is motivated by the pressing need for affordable, real-time, and accurate WQM and WQP systems in aquaculture. The research consists of interconnected works, each addressing specific aspects of WQM and WQP using DL methodologies. The first part of the thesis focuses on developing a cost-effective virtual sensing (VS) system for real-time WQM. Traditional methods relying on expensive physical sensors are impractical for small-scale farms. To overcome this challenge, the proposed deep learning enhanced virtual sensing (DLEVS) system leverages DL techniques to significantly improve the accuracy of the VS-based WQM system, reducing reliance on costly sensors. In the second work, LSTM, GRU, and their bidirectional variants are explored for precise WQP. Accurate prediction of WQ is crucial for effective aquaculture management. By leveraging the non-linear modeling capabilities of DL, these models aim to enhance prediction accuracy and provide valuable insights into future WQ conditions.

The third work reported in this thesis focuses on refining the accuracy and compu-

tational efficiency of WQP through the proposed hybrid DL models. Hyperparameter optimization is extensively studied to maximize model performance while minimizing computational requirements, facilitating real-time implementation. The fourth work introduces attention-based DL models to further enhance the accuracy and computational efficiency of WQP. These models employ attention mechanisms to concentrate on the most relevant features and parameters, resulting in improved prediction performance. A comprehensive study of model performance with different hyperparameters is conducted to identify the optimal configuration that balances accuracy and computational resources.

The research works reported in this thesis contribute to the integration of advanced DL techniques into WQ management in aquaculture. By addressing challenges related to cost-effectiveness, prediction accuracy, and computational efficiency, this research work enhances the sustainability of aquaculture.

Keywords: Aquaculture, IoT, virtual sensing, water quality monitoring, water quality prediction, deep learning, attention-based models.